

Mutual Independence

Recall the definition of *mutual independence* among a set of n random variables:

DEF 4.6.5 (Mutual Independence) Let $\mathbf{X} = (X_1, \dots, X_n)'$ be an n -dimensional random vector with joint pdf or pmf $f_{\mathbf{X}}(x_1, \dots, x_n)$ and marginal pdfs or pmfs for each X_i denoted by $f_{X_i}(x_i)$. Then $X_1, \dots, X_n$ are **mutually independent random variables** if, for every $(x_1, \dots, x_n)' \in \mathbb{R}^n$,

$$f(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) = \prod_{i=1}^n f_{X_i}(x_i).$$

Since the equation above holds for all $(x_1, \dots, x_n)' \in \mathbb{R}^n$, this definition implies that any subset of these random variables are also mutually independent. For example, let $k < n$ and consider the joint pdf/pmf of $(X_1, \dots, X_k)'$:

$$\begin{aligned} f(x_1, x_2, \dots, x_k) &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_{k+1} \cdots dx_n \\ &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_1}(x_1) \cdots f_{X_n}(x_n) dx_{k+1} \cdots dx_n \quad \text{by independence} \\ &= f_{X_1}(x_1) \cdots f_{X_k}(x_k) \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_{k+1}}(x_{k+1}) \cdots f_{X_n}(x_n) dx_{k+1} \cdots dx_n \\ &= f_{X_1}(x_1) \cdots f_{X_k}(x_k) \int_{-\infty}^{\infty} f_{X_n}(x_n) \left(\int_{-\infty}^{\infty} f_{X_{n-1}}(x_{n-1}) \cdots \left(\int_{-\infty}^{\infty} f_{X_{k+1}}(x_{k+1}) dx_{k+1} \right) \cdots dx_{n-1} \right) dx_n \\ &= f_{X_1}(x_1) \cdots f_{X_k}(x_k) \prod_{i=k+1}^n \left(\int_{-\infty}^{\infty} f_{X_i}(x_i) dx_i \right) \\ &= f_{X_1}(x_1) \cdots f_{X_k}(x_k) \quad \text{since each pdf integrates to 1} \end{aligned}$$

where integrals are replaced by sums in the discrete case.

Therefore, if $X_1, \dots, X_n$ are mutually independent, then any subset $\{X_{j_1}, \dots, X_{j_m}\}$ of random variables are also mutually independent, where $m \in \{2, \dots, n\}$ and $j_i \in \{1, \dots, n\}$ for all $i = 1, \dots, m$.